Two definitions of young give opposite answers in a fibroblast rejuvenation screen

In one public screen, the same four transcription factors move cells toward early-passage cells and away from young human donors. The difference is what each target calls young.

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SUMMARY

A rejuvenation screen needs a definition of young before it can call anything a hit. Sengstack et al. perturbed 200 transcription factors in dermal fibroblasts aged by repeated passaging, and took early-passage cells as young. For 340 of their perturbations, we asked whether each one moves cells closer to young cells. We used two definitions of young: the authors' early-passage cells, and fibroblasts from young human donors in the Genotype-Tissue Expression project (GTEx). We first checked that the test could detect an effect of known size. The two definitions give opposite answers. All four factors the authors validated (E2F3, EZH2, STAT3 and ZFX) move cells closer to early-passage cells, and all four move them away from young donors. Across the screen, only 5 of 340 perturbations are called closer to young donors, fewer than the 17 expected by chance. The reason lies in the aging model. Passaging ages cells by making them divide until they stop. In this dataset, the direction from late to early passage points almost the same way as cell division measured in the cells themselves, and runs nearly perpendicular to donor age. A screen that calls early-passage cells young therefore cannot tell perturbations that restore division from perturbations that make cells young. A rejuvenation claim needs a young target defined by donor age, measured on the same platform as the perturbations. This is a point about the readout, not the factors; the authors' cellular and in vivo results lie outside this analysis.

INTRODUCTION

Most rejuvenation screens define young through a signature: genes that change between old and young cells, and a score that rewards perturbations reversing that change. The score's verdict can be no better than the signature's choice of old and young.

Sengstack et al.[1] perturbed 200 transcription factors by CRISPR activation and interference, 400 perturbations in all, in passaged neonatal dermal fibroblasts. Their aging model is replicative[2]: cells passaged beyond 31 population doublings (PD) are old, and wild-type cells at PD14 are young. A perturbation is scored by R_{rej}, the correlation between two sets of log2 fold changes, late versus early passage and perturbation versus non-targeting control; factors with R_{rej} of −0.3 or below were taken as top hits. Four were validated with phenotype data: E2F3 and EZH2 by overexpression, STAT3 and ZFX by repression.

A correlation of this kind records the direction in which cells moved, not where they ended up. A companion preprint[3] showed that directional readouts cannot tell an approach to young cells from a departure from old ones. It proposed a replacement: the distance from a sample's expression profile to a young target of the same cell type, measured across thousands of genes, and checked before any result is read against an effect of known size and against the noise at the experiment's own depth.

Here we apply that readout to the Sengstack data twice: once with the authors' own young target, and once with young human donors. The two targets give opposite answers for the same four factors, and we trace the reason to the aging model. The authors argue, on four lines of evidence, that their factors act beyond proliferation; we take each in turn in the Discussion.

RESULTS

What the test measures, and what it can detect

The public deposit of Sengstack et al.[1] holds 36,514 cells from one neonatal dermal fibroblast line: wild-type cells at three passages, and two CRISPR screens, each with its own non-targeting control cells (Figure 1). Each population appears in only one of these groups, so the young target never shares a batch with a perturbation. Of the 400 perturbations, 340 have at least 30 cells and are analyzed. Four other public datasets serve as references: GTEx v10[4] cultured fibroblasts from 652 donors aged 20–79 years, which give the young- and old-donor targets and the compressed coordinate space described below; the Southard[5] and Lu[6] datasets, which give independent measures of cell division and a second donor-age direction; and the companion preprint[3], which supplies the reference gene set of 23,485 genes and a frozen age ruler, a ridge regression on donor age trained on the GTEx fibroblasts. Cell counts, guide assignment and the reference datasets in full are in Supplementary Note S1 and the Methods.

Each group of cells is summarized as one expression profile: the pooled, normalized expression of every gene, scaled against GTEx fibroblasts, so that a profile is a point in a space with one axis per gene. Distance is the ordinary straight-line (Euclidean) distance between two such points, so it adds up the differences across all genes. A perturbation's effect is the difference between its own pool of cells and the cells of the same screen that received a non-targeting guide, which stand in for the unperturbed state. We add that effect to a starting profile and ask whether the result lies closer to the young target than the starting profile did. Small groups of cells also move by chance, so each perturbation is compared with 200 random samples of control cells of the same size. It is called closer when its change in distance is negative and smaller than that of 95% of those samples; that value is its threshold. About 5% of perturbations, 17 of 340, are therefore expected to be called by

chance. Both starting profiles were chosen before any result was computed. The main one is the perturbation's own control cells (the own-control start), which share its screen's genetic background. The second is late-passage wild-type cells at PD32 (the late-passage start), which lie on the passage series itself and so carry none of the offset between the screens and the wild-type cells. The two agree on all four factors; results from the second are in the supplement. The young-donor test later in this section starts from a third profile, the average old donor.

Distance can be measured in two coordinate spaces. The full gene space is the one just described, with one axis for each of 23,485 reference genes, of which 16,781 are measured in this deposit. With so many axes, sampling noise in every gene adds to the distance. The 20-component space instead describes each profile by its position along the 20 directions in which GTEx fibroblasts from middle-aged donors (40–59 years) differ most, found by principal component analysis; the young and old target donors and every Sengstack cell lie outside that fit. The number of components was carried over from an earlier analysis of the Southard screen[5], where 20 detected a planted effect most reliably; it was not tuned on these data (Methods). Compressing this way shortens the distance between late- and early-passage cells to 45.1% of its length in the full gene space, but discards a far larger share of the sampling noise. That trade is why the test can register an effect here and not there.

Before reading any factor, we ran the test on control samples carrying no effect and on samples carrying a planted effect of known size (Supplementary Figure S1). In the full gene space the test cannot call anything at the four factors' group sizes (96 to 157 cells): sampling noise alone moves every control sample away from the target, and a planted effect of half the passage gap is detected in 0.000 to 0.915 of samples. In the 20-component space the control samples are centered on zero, and the test wrongly calls 0.040 to 0.085 of them, against 0.05 expected and a limit of 0.10 set in advance. Planted effects of half the passage gap are detected with power 0.940 to 1.000, and effects of a quarter of the gap with 0.565 to 0.740. Results are therefore reported in the 20-component space. They can be trusted for effects at least half as large as the whole late-to-early passage difference; a smaller real effect could be missed. In the full gene space only ZFX among the four factors, and 3 of 340 perturbations, were called; that near-empty result reflects the space's insensitivity at this depth, not the factors.

Two more things had to be checked before the test could be trusted. First, the cells carrying perturbations are engineered CRISPRa and CRISPRi derivatives, not the wild-type cells that define early and late passage, so moving from a screen's control cells toward PD14 is not purely a move back along the passage series. The gap between the two populations is about as long as the passage gap itself (0.65 to 1.05 times), but it runs roughly at right angles to the passage direction: 84–86° in the full gene space and 100–107° in the 20-component space. It therefore adds a sideways component without cancelling the passage step, and the line from a screen's control cells to PD14 still points mostly along passage (cosine 0.81–0.89 in the 20-component space and 0.65–0.79 in the full gene space; Supplementary Note S1, Supplementary Figure S2). Second, passage is not donor aging. The passage direction is nearly perpendicular to the direction in which GTEx fibroblasts differ with donor age: the cosine between them is −0.040 in the full gene space and −0.199 in the 20-component space, where 0 would mean exactly perpendicular and 1 the same direction.

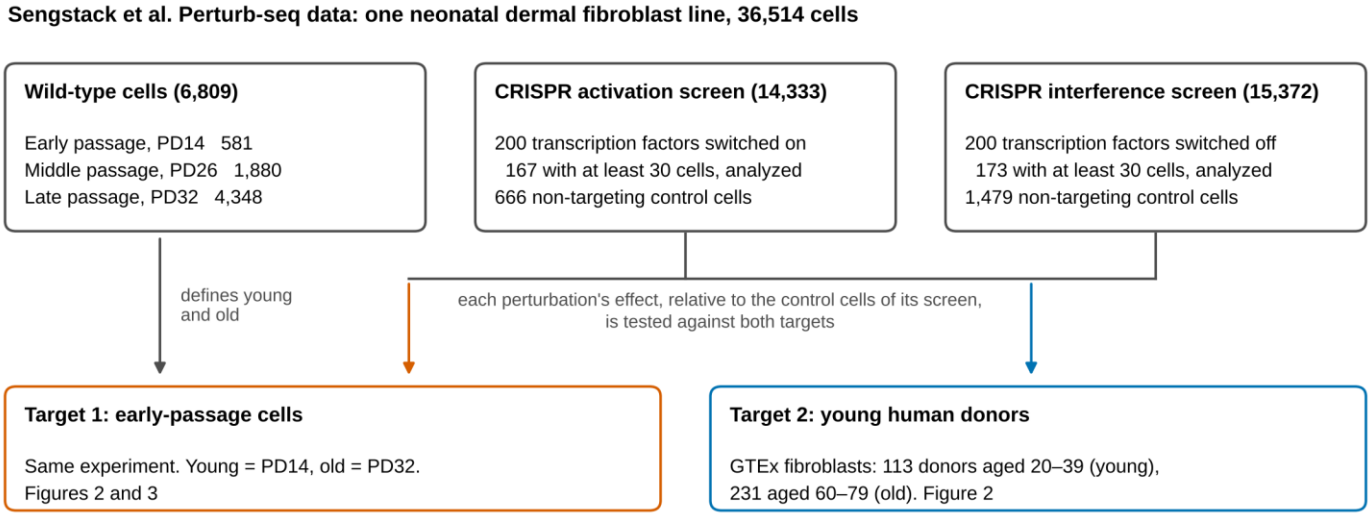


Figure 1. Study design. The Sengstack et al.[1] deposit holds wild-type cells at three passages and two CRISPR screens, each with its own non-targeting control cells. Early (PD14) and late (PD32) passage cells define the first target. GTEx[4] fibroblasts from young (20–39 years) and old (60–79 years) donors define the second. Each perturbation's effect, relative to the control cells of its own screen, is tested against both targets. Orange marks the early-passage target and blue the young-donor target throughout the paper.

Against early-passage cells, all four approach

In the 20-component space, 80 of 340 perturbations are called closer to early-passage cells, against 17 expected by chance (Figure 2, orange). All four validated factors are among them: change in distance −19.2 (E2F3), −13.2 (EZH2), −10.3 (STAT3) and −19.9 (ZFX), against thresholds of −5.9 to −7.2. Another 111 perturbations end closer without passing their threshold; with no effect at all, about half of all perturbations would land on that side by chance, which is why each must pass a threshold. The test is run on every perturbation, but we interpret only the four factors that Sengstack et al. validated, not the other 76 called. From the late-passage start, 73 are called, again including all four.

Perturbations with more cells are easier to call, but that does not

account for the four. The call rate — the fraction of perturbations called closer — rises, unevenly, from 0.056 in the tenth of perturbations with the fewest cells to 0.545 in the tenth with the most, and the four factors sit at the 82nd to 99th percentile of cell count (Supplementary Figure S2C). The comparison that holds cell count fixed is within a band of similar size: among the 60 perturbations closest to the four in cell count, 24 are called, and the four are still among the strongest movers in that band, at the 1.7th, 3.3rd, 15.0th and 21.7th percentile of change in distance.

Against young donors, all four move away

We then asked the question the passage model cannot answer: does each factor move cells toward young human donors? Each perturbation's effect, measured against its own control cells, was added to the average profile of GTEx donors aged 60–79, and the test asked whether the result lay closer to the average of donors aged 20–39, with thresholds built as before. The effect is the same vector as in the passage test; only the starting profile and the target change. At the four factors' group sizes this test detects a planted effect of a quarter of the donor-age gap with power 0.945 to 0.990.

All four move away (Figure 2). Their change in distance to young donors is +11.4 (E2F3), +9.0 (EZH2), +5.3 (STAT3) and +15.3 (ZFX), against thresholds of −1.0 to −1.3. Each effect points away from the donor-age direction (cosines −0.14 to −0.27), along the passage direction (+0.72 to +0.78), and 26–38° from its screen's cell-division direction. Across the screen, only 17 of 340 perturbations end any closer to young donors, and 5 are called, fewer than the 17 expected by chance. None of the 80 perturbations called closer to early-passage cells is called closer to young donors, and all 80 move away from them. Figure 2 plots each change relative to its own threshold, because the threshold depends on group size: a perturbation with many cells can move further than one with few and still fall short of its stricter threshold.

Against this target, distance adds little to direction. Each effect starts from the old-donor average, so any effect that points away from the donor-age direction, as all four do, must increase the distance to young donors. Distance would add information only for an effect that pointed toward young donors, where it could tell an approach from an overshoot. Separately, the four genes are also uncalled and move away from young donors in the Southard screen[5], a different neonatal line; that screen used activation only, so for STAT3 and ZFX it tests the opposite perturbation.

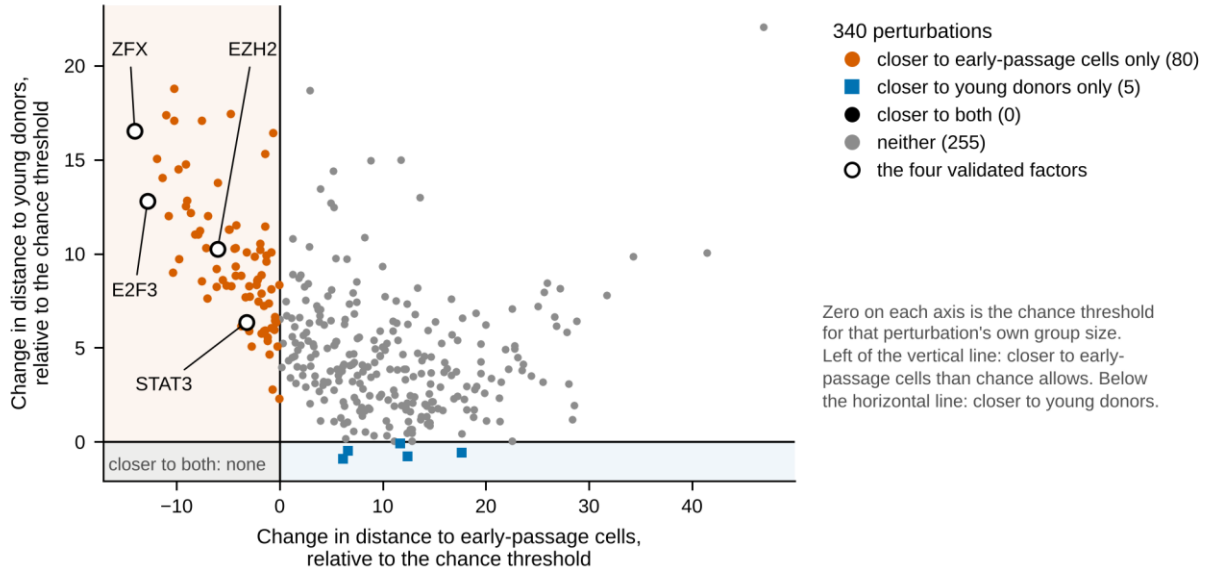


Figure 2. The two targets give opposite answers. Each of the 340 perturbations, plotted by its change in distance to early-passage cells (x) and to young donors (y), each relative to that perturbation's own chance threshold, in the 20-component space. Zero is the threshold on both axes, so points left of the vertical line are called closer to early-passage cells (orange) and points below the horizontal line are called closer to young donors (blue). No perturbation is called closer to both. The 80 called closer to early-passage cells, including the four validated factors, all lie above the horizontal line.

That approach tracks cell division

Across the screen, a perturbation's change in distance to early-passage cells closely tracks its effect on cell-cycle genes (Spearman $\rho = -0.864$; Figure 3C). The cell-cycle score is the mean expression of the S and G2/M genes of Tirosh et al.[7] relative to the screen's control cells, so a higher score means more cells in active division. Because the score and the starting profile use the same control cells, this correlation is descriptive rather than a test. Three of the four factors are at or above the 90th percentile of the score among perturbations of similar size, and 79 of the 80 called perturbations raise the score, against 199 of all 340.

We then asked whether the passage signal is anything more than cell division. If it is, removing the cell-division direction from the data should leave an approach toward early-passage cells that the test can still detect. A cell-division direction is built by scoring each cell for cell-cycle activity with the S and G2/M genes of Tirosh et al.[7] and taking the difference between the quarter of cells with the highest score and the quarter with the lowest. We chose three such directions before running the test, and named one as primary.

The primary direction, from a screen's own control cells, lies 19.7° and 20.6° from passage in the 20-component space: the two point almost the same way. Removing it therefore also removes about two-thirds of the passage direction, and the test can no longer detect even an effect we planted on purpose (power falls from 0.895

to 0.010; Figure 3B). Cell division cannot be removed without removing passage, and by the criteria we set in advance the two are inseparable in this dataset. Called perturbations fall from 82 to 27, with two of the four factors still called, but a test that cannot see a known effect cannot show that anything is missing, so this drop is evidence neither for nor against rejuvenation.

The second direction, built from middle-passage PD26 wild-type cells that neither screen uses, lies 18.5° from passage and gives the same picture, so the result does not depend on the screens' engineered cells. The third is cruder: the cell-cycle gene list itself, with equal weight on each gene and no cells. It lies 61.7° from passage, so removing it takes away only 12.0% of the passage direction; the test still works (power at least 0.885), and all four factors stay called. This shows that removal itself does not break the test; it breaks only when the removed direction is nearly parallel to passage. It also means the conclusion holds for cell division measured in cells, not for this cruder measure (Supplementary Note S2).

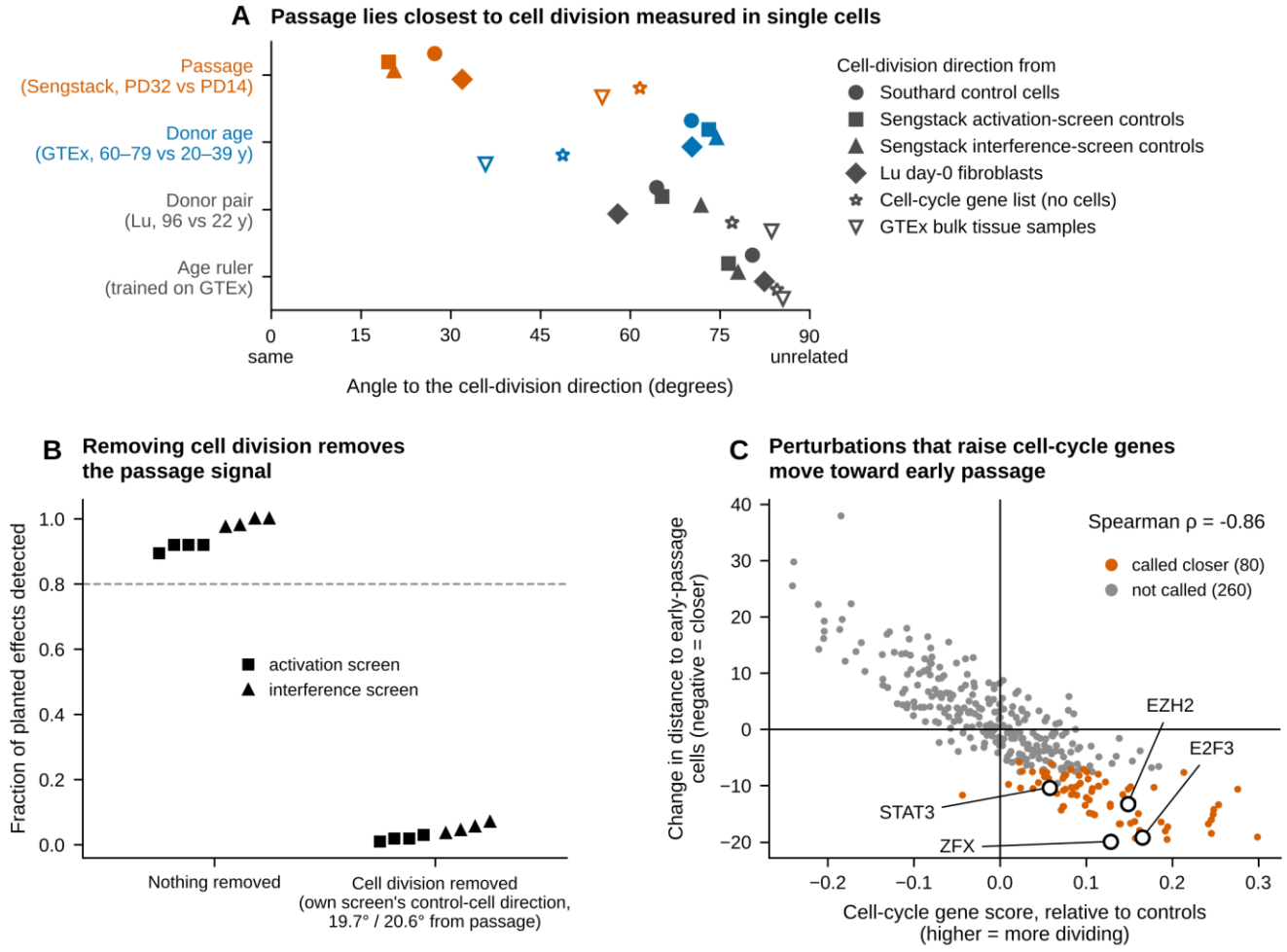


Figure 3. Passage and cell division point in nearly the same direction in this dataset, and cannot be separated here. (A) Descriptive comparison: angle between four aging directions and cell-division directions, in the 20-component space; 0° means the same direction and 90° unrelated. Filled symbols, directions built from single cells of three datasets; open symbols, directions not built from single cells (Table 1). Values in the full gene space are in Table 1. (B) Fraction of planted half-gap effects detected from the own-control start with nothing removed and after removing each screen's own control-cell direction, which lies 19.7° (activation) and 20.6° (interference) from passage. Squares, planted effects in the activation screen; triangles, in the interference screen; each point is one group size tested (96, 107, 128 and 157 cells, the group sizes of the four validated factors), with 200 planted samples each. Dashed line, 0.80. (C) Each of the 340 perturbations: change in cell-cycle gene score relative to its control cells against change in distance to early-passage cells. Orange, called closer: below the perturbation's own threshold, which ranges from −4.8 to −12.2 with group size, so the line at zero is not the calling line. Circled, the four validated factors.

In single-cell measures, passage lies closest to cell division

We also asked whether passage is unusual in lying so close to cell division. We compared four aging directions — passage, donor age in GTEx, the frozen age ruler of the companion preprint[3], and a 96- versus 22-year-old donor pair from Lu et al.[6] — with the cell-division directions of Table 1 (Figure 3A). The main argument does not depend on this comparison. It rests on two other results: passage cannot be separated from cell division measured in cells, and passage is nearly perpendicular to donor age.

The test we planned compared donor age and passage, each against a cell-division direction built from its own samples: bulk GTEx tissue for donor age, single cells for passage. Its result fell in the intermediate range we had defined in advance. Donor age lay 35.8° from cell division, against 19.7° for passage, so donor age is the better separated of the two; but removing cell division would still take away 41.5% of the donor-age direction, and the bootstrap interval of that angle is wide (20–70°).

Without a planned test, we also compared every aging direction

with every cell-division direction built from single cells. Each of them placed passage closest to cell division: 20–32° in the 20-component space, against 58–82° for the other three definitions, and 53–71° against 75–89° in the full gene space. Where bootstrap intervals were computed, they do not overlap: passage lies 18–24° from the Sengstack control-cell direction, donor age 56–90°. The pattern does not hold for the two directions not built from single cells; with either of them, donor age is the closer definition (open symbols in Figure 3A; Supplementary Note S3). The bulk GTEx direction points strongly toward the old donors (cosine −0.81 with donor age). We suspect it reflects how tissue was sampled rather than cell division, but this explanation is post hoc and untested. Taken together, passage lies closer to cell division than donor age does whenever cell division is measured in cells, and the two constructions that disagree are the least direct measures of dividing cells. We read this as a real feature of the passage model, though no test planned in advance confirms it.

Table 1. The six cell-division directions compared, and how close each definition of aging lies to them. Each direction runs from the quarter of units with the lowest cell-cycle score (S and G2/M genes of Tirosh et al.) to the quarter with the highest; the gene-list direction instead puts equal weight on each of the 94 genes and uses no cells. Angle in degrees (0°, same direction; 90°, unrelated); 20-component space first, full gene space in brackets. The two spaces are not directly comparable: unrelated directions sit about 80° apart in 20 dimensions and nearly 90° in 23,485. In every row built from single cells, passage lies closest to cell division; in the two rows not built from single cells, donor age does. The Sengstack control-cell directions use one half of each screen's control cells. Lu values are corrected for a gene-matching error noted in the repository. A further direction, from middle-passage PD26 cells, was built for the removal test only (18.5° from passage). Descriptive only.

Cell-division direction (one per row), built from	Cells	Angle to each definition of aging (one per column)				Used in
		Passage (Sengstack)	Donor age (GTEx)	Donor pair (Lu, 96 vs 22 y)	Age ruler (trained on GTEx)	
Built from single cells						
Southard control cells	14,013	**27.3° (66.0°)**	70.2° (82.5°)	64.5° (80.1°)	80.5° (88.8°)	Fig 3A
Sengstack activation-screen control cells	333 (half)	**19.7° (59.8°)**	73.2° (85.6°)	65.4° (81.0°)	76.4° (89.1°)	Fig 3A, 3B
Sengstack interference-screen control cells	739 (half)	**20.6° (52.8°)**	74.4° (84.8°)	71.8° (81.6°)	78.0° (89.0°)	Fig 3A, 3B
Lu day-0 fibroblasts	12,803	**31.9° (70.9°)**	70.4° (82.4°)	58.0° (74.7°)	82.5° (88.7°)	Fig 3A
Not built from single cells						
Cell-cycle gene list, equal weights	none (94 genes)	**61.7° (87.1°)**	48.8° (85.0°)	77.1° (88.4°)	84.6° (89.1°)	Fig 3A
GTEx bulk tissue samples	652 donors	**55.4° (75.0°)**	35.8° (45.1°)	83.7° (88.2°)	85.5° (89.6°)	Fig 3A

A directional score calls rejuvenation where distance finds none

On the frozen age ruler from the companion preprint[3], 181 of 340 perturbations score younger. Of those, 116 are not called closer to early-passage cells in the 20-component space (120 from the late-passage start). This is a diagnostic, not evidence about any factor, and it repeats the companion preprint's finding in a further dataset (Supplementary Figure S4; late-passage start, Supplementary Figure S3E).

DISCUSSION

Passage number models replicative exhaustion, not donor aging. Replicative aging is defined by the loss of division capacity[2], so in a passage series one would expect the direction from old to young to be largely the direction of renewed division. Here the passage direction lies about 20° from cell division measured in the screen's own control cells, and nearly perpendicular to donor age. Whenever cell division is measured in cells, passage lies closer to it than the other definitions of aging we examined. Removing the screen's own cell-division direction removes the passage signal with it, so within this dataset restored youth and restored proliferation cannot be told apart. This is a property of the model, not a flaw in anyone's analysis, and it limits any reanalysis of these data, including ours.

The two targets give two answers and one picture. The four validated factors make cells resemble early-passage cells, and their effects lie close to the cell-division direction. The same effects carry cells away from young human donors. Both statements hold at once because "young" means two different things in the same dataset.

The authors address proliferation directly[1]. They state that "most of the top TF perturbations up-regulate cell cycle–related TF modules", but also that "many top TF perturbations also reverse gene expression changes … not obviously attributable to cell cycle redistribution", shown by R_{rej} computed within a single cell-cycle phase, and they conclude that the perturbations "reverse the aspects of aging that are not directly related to cell proliferation". They report the same transcriptional signature in skin fibroblasts from young and old patients, and in tissues of aged mice rejuvenated by parabiosis, where cells showing it are enriched for mitotically active cell types; they find that the factors' effects bear no resemblance to cancer signatures, in vitro and in liver; and EZH2 overexpression in aged mouse liver reversed aging-associated expression (R_{rej} −0.42) with reduced steatosis and fibrosis and improved glucose tolerance. They also note that the slowdown of cell-cycle progression, a major hallmark of aging in vitro, "may not be an important feature of in vivo aging".

Each of these four lines of evidence deserves an answer. First, R_{rej} survives within a single cell-cycle phase. That removes redistribution between phases, but it is still a correlation of fold changes, so it records the direction of change and not where cells end up, and proliferation-associated programs need not be confined to phase membership. Our cell-division direction is built from a cell-cycle score, not from phase, so this analysis neither shows nor excludes a within-phase component. Second, the signature appears in fibroblasts from young versus old patients. On direction alone, that

finding and ours disagree: here the four effects point away from the GTEx donor-age direction (cosines −0.14 to −0.27). Why they differ is unclear. Their comparison uses transcription-factor modules pre-selected as differential between PD14 and PD32, in a different donor dataset; ours uses all measured genes against GTEx. Separately, the enrichment for mitotically active cell types they report comes from aged mice rejuvenated by parabiosis, and is also consistent with a proliferative component. Third, EZH2 improved aged mouse liver. A screen with a confounded readout can still return factors that work, and that result is a functional test outside anything transcriptome distance can judge; it is the right kind of evidence, and nothing here weakens it. Fourth, the factors' effects bear no resemblance to the expression changes of cancerous transformation. That argues against tumorigenicity, but not against a proliferation program: the cancer signatures they compare with also carry stress-response, adhesion, migration and metabolic changes, so a perturbation can differ from them and still act mainly through renewed division. The authors also report that the factors did not generally change methylation age, which they attribute to senescence being unrelated to epigenetic clocks; this analysis does not bear on it.

One lesson is general. Measured across all genes at about 100 cells per factor, the test could not have called anything: even control samples with no effect never ended closer to the target, so the near-empty result there carries no information (Supplementary Figure S1). Planting an effect of known size at the screen's own depth, before reading any factor, showed which space could see an effect and which could not. Any readout of this kind, including a correlation, should be checked the same way.

What is not claimed. Nothing here concerns aged human cells, because the substrate is one neonatal line throughout. Transcriptome distance does not test the authors' proliferation, proteostasis, mitochondrial or lysosomal phenotypes. No factor is recommended. The claim is about what a passage-anchored readout can resolve: in this dataset it cannot tell renewed division from rejuvenation.

The remedy is general. A rejuvenation screen needs a young target that is young by donor age, measured on the same platform as the perturbations, ideally with cells from several old and young donors in the same run. With such a target, approach toward youth and renewed division are no longer measured along the same direction, so the two have a chance to be told apart.

LIMITATIONS

- One neonatal fibroblast line, with passage as the only age variable. The CRISPRa and CRISPRi cells are engineered derivatives of that line, treated here as the same line as the wild-type passage arms, which is an assumption, not a measurement; they also carry no PD label and may have drifted by about three population doublings during the experiment.
- The young target sits in a different sample group from every perturbation. The offset between the screens and the wild-type cells was measured, not removed. It is an upper bound on the technical difference between them, because the screen cells are known only to be past 31 population doublings, not to be at PD32, so any real passage difference is mixed into it. Depth matching by thinning is weak, because log2 counts per million already divide out depth, so the depth difference between sample groups is reported, not removed.
- Angles depend on the coordinate space: passage and cell division are 19.7° apart in the 20-component space and 59.8° in the full gene space (activation screen). The criteria for separability were set in the 20-component space. For directions built from single cells, the ordering of aging definitions holds in both spaces, but the size of the gap does not.
- The 20-component space keeps 45.1% of the passage gap's length, and its number of components was set on another dataset and not varied here. Its false-call rate on control samples was 0.085 from the own-control start, under the limit of 0.10 set in advance, but 0.110 from the late-passage start, which would have exceeded it. The own-control start was chosen in advance for the power check; using it for the false-call check as well was written down after the 0.110 figure was known, and that choice should be read in that light.
- The passage test detects half-gap effects (power at least 0.940) but not reliably quarter-gap ones (at least 0.565). The call rate rises with cell count, and the four factors are among the perturbations with the most cells.
- The passage direction and the target are built from the same PD14 and PD32 cells, and the z-scoring constants come from all 652 GTEx donors, including the anchors.
- The GTEx-anchored test adds an effect measured in single cells to a bulk-tissue starting profile. Late-passage Sengstack cells sit 14.4 and 14.3 times the donor-age gap from the young and old GTEx averages in the full gene space, and 22.0% of the donor-age direction's squared length lies in genes this dataset does not measure. Against this target, a departure follows from the sign of each effect's cosine.
- The comparison of aging definitions (Figure 3A, Table 1) is descriptive. Its planned test gave an intermediate result with a wide interval, and the pattern holds only for cell-division directions built from single cells. The angles in Table 1 are shown without confidence intervals.
- The inseparability result rests on the primary cell-division direction. Of the other two, the one built from cells agrees with it; the gene-list direction, which lies far from passage, does not (Results).

METHODS

Data. The Perturb-seq deposit of Sengstack et al.[1] is one h5ad file (figshare 10.6084/m9.figshare.30898748, CC BY 4.0; md5 bfccaa94a916c22e4b0793913a04128b). It holds 36,514 cells by 36,601 genes as raw counts. GTEx v10[4] cultured fibroblasts (652 donors) are the fixed reference throughout. The frozen age ruler, the GTEx young (20–39 years, 113 donors) and old (60–79, 231 donors) average profiles and the Lu donor-pair axis are read unchanged from the companion preprint's archive[3].

Pre-registration. Four successive analyses of these data, after an inventory of the file that computed no statistic, were each specified in a timestamped flag file before any of their numbers existed: statistic, nulls, positive controls, thresholds and the wording of every possible conclusion. They ran in sequence; the second was preregistered after the first's sensitivity check had failed. Ambiguities found before computing were resolved in writing in an amendments file beside each flag; one amendment that bears on a decision rule was written with a number known, and is disclosed under

Limitations. Only the reading whose conditions were met is reported as a conclusion; everything else is labeled descriptive. An earlier pre-registered analysis of these data (September 2026) used a directional cosine. Its pre-registered entry check required at least three of the four named factors to align with the passage direction at a false-discovery-rate-adjusted $q \leq 0.05$; all four pointed in the published direction but only one reached significance, so the check failed and no factor-level result was interpreted. That analysis is superseded, retained in the archive with a correction notice, and not used here. These are self-administered protocols recorded in the repository, not deposits with an external registry.

Cells, assignment and pseudobulks. Wild-type passage arms were taken from the PD14, PD26 and PD32 flags; perturbed cells carry no PD label. Guide columns are named by target, screen and replicate (two per perturbation, except DLX5 interference and PHOX2A activation with one). A cell was assigned to the guide whose count was a unique strict maximum, and a cell whose guide was non-targeting became a control cell of the screen given by the file's activation or interference flag; 2,680 cells with a tied maximum and 270 whose winning guide disagreed with the file's own modality label were left unassigned. Perturbations with fewer than 30 assigned cells were excluded, leaving 340 of 400. Counts were summed per group, TMM-normalized[8], converted to log2 counts per million (prior count 2) and z-scored with the frozen GTEx means and standard deviations. The file carries gene symbols only, so 16,781 of the 23,485 ruler genes were matched by symbol; unmatched genes are set to zero. Where sequencing depth was compared, groups were thinned binomially to the lowest group median (200 repeats).

Coordinate spaces. The full space is all 23,485 reference genes of the companion preprint[3]. The compressed space projects differences onto the first 20 principal components of the cultured fibroblasts of 308 GTEx donors aged 40–59. The number of components was fixed in an earlier pre-registered analysis of the Southard screen[5], which compared 20, 50 and 100 components and made 20 primary because it recovered a planted effect of a quarter of the gap most reliably at that screen's depth of 106 cells (power 0.990, against 0.965 and 0.930; the full gene space reached 0.115). Those powers were measured with a basis fit on all 652 GTEx donors; the basis was then refit on the 308 middle-aged donors to keep the target donors out of the fit. It is read frozen here; 50 and 100 components were not run on these data.

The approach test. The passage direction is PD14 minus PD32, each a random half (half A) of its cells; the target is PD14 half A, and PD14 half B is reserved for the mixing control. A perturbation's effect is its pseudobulk minus its own screen's control-cell pseudobulk. The effect is added to two starting profiles fixed in advance: the control-cell pseudobulk itself (own-control start, primary) and PD32 half A (late-passage start), which removes the offset between groups. The statistic is the change in Euclidean distance to the target. The threshold is the 5th percentile of the same statistic for 200 random samples of control cells from the same screen at the perturbation's own cell count. A perturbation is called closer when its change is negative and below its threshold; 17 of 340 are expected by chance. Figure 2 plots each change minus its threshold, so that zero is the threshold on both axes; this is arithmetic on stored values and changes no call.

Checks run before any factor result. False-call rate: each screen's control cells were split in half; thresholds were built on one half and the test run on 200 control samples with no effect drawn from the other. Planted effects: control samples were displaced along the passage direction by 25, 50 and 100% of its length; power is the fraction of samples the test calls. False-call and planted-effect checks were run at the four factors' group sizes (96, 107, 128 and 157 cells) in both screens. Mixtures: PD14 half B mixed into control samples at 10, 25 and 50% (96 and 157 cells, both screens) moved them toward the target by 0.087–0.105, 0.215–0.248 and 0.433–0.484 of the gap (medians; 20-component space, own-control start).

Cell-division directions. Each is the difference between the top and bottom quartile of units ranked by a cell-cycle score (mean log1p of counts per 10,000 over the S and G2/M genes of Tirosh et al.[7], 42 and 52 present). Units are one half of each Sengstack screen's non-targeting control cells (333 activation, 739 interference), which enter no starting profile, target or threshold; Sengstack PD26 wild-type cells (1,880); Southard non-targeting cells (14,013)[5]; Lu day-0 fibroblasts (12,803)[6]; and GTEx donors (bulk tissue, 652). The gene-list direction has weight 1 on each of the 94 cell-cycle genes in the reference gene set and 0 elsewhere, scaled to unit length. Angle from parallel is $\arccos|\cos|$, so that 0° means the same direction and 90° unrelated; the share of a direction removed by projecting a direction out is $1 - \sqrt{(1 - \cos^2)}$. Angles are reported per space and never compared across spaces.

Young-donor test. Each effect, measured within one screen, was added to the average profile of GTEx donors aged 60–79 and scored by its change in distance to the average of donors aged 20–39 in the compressed space, with thresholds from control samples as above. Planted effects were displaced along the young-minus-old donor direction.

Resampling and figures. Seeds 20260914 (splits, nulls, permutations) and 20260918 (bootstraps); 200 repeats throughout; intervals are 2.5th–97.5th percentiles. Figures were drawn from stored result files without recomputation.

DECLARATIONS

Data and code availability. All input data are public: figshare 10.6084/m9.figshare.30898748 (Sengstack et al.[1]) and GTEx v10[4]. Code, pre-registration flags, amendments, correction notices and result files are available at https://github.com/k-dlrgus/rejuvenation-readouts and archived at https://doi.org/10.5281/zenodo.23591315.

Author contributions. R.K. is the sole author. He conceived the study, specified the analyses and their pre-registered decision rules, directed their execution, and is responsible for the manuscript.

Use of AI tools. Analysis code was written by AI coding agents under the author's written pre-registration protocols, and the manuscript was drafted with AI assistance. The author reviewed all analyses and takes full responsibility.

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SUPPLEMENTARY NOTES

Supplementary Note S1. The data, the reference datasets, and the boundary between the screens and the wild-type cells

The public deposit of Sengstack et al.[1] holds 36,514 cells from one neonatal dermal fibroblast line (Figure 1). Wild-type cells at three passages (PD14, 581 cells; PD26, 1,880; PD32, 4,348) form one group, and the CRISPR activation (14,333 cells) and interference (15,372) screens form two more. The screens use engineered CRISPRa and CRISPRi derivatives of the line, and no population appears in more than one group, so the young target lies outside the batch of every perturbation. For each cell the deposit records the count of each of 804 guides (798 targeting, two per perturbation with two exceptions, and 6 non-targeting) but not which guide the cell received. We assigned each cell to the guide with the highest count when that maximum was unique (Methods). Cells whose top guide is non-targeting are the control cells of their screen, which is read from the file's own activation or interference label: 666 in the activation screen and 1,479 in the interference screen. Of the 400 perturbations, 340 have at least 30 cells and are analyzed (167 activation, 173 interference).

Four other public datasets serve as references. GTEx v10[4] provides bulk RNA-seq of cultured skin fibroblasts from 652 donors aged 20–79 years: the average profiles of donors aged 20–39 (113 donors) and 60–79 (231) are the young and old targets for donor age, and donors aged 40–59 (308) define the compressed coordinate space described in the Results. Southard et al.[5] screened transcription factors by CRISPR activation in a different neonatal fibroblast line; their 14,013 non-targeting control cells give an independent measure of cell division. Lu et al.[6] profiled dermal fibroblasts from a 96-year-old and a 22-year-old donor before and during partial reprogramming; the untreated (day-0) cells give a second donor-age direction, from the older to the younger donor, and, pooled (12,803 cells), a further measure of cell division. The companion preprint[3] supplies the reference gene set (23,485 genes) and a frozen age ruler, a ridge regression on donor age trained on the GTEx fibroblasts. Cell-cycle activity is scored with the S-phase and G2/M gene lists of Tirosh et al.[7]

The boundary between the screens and the wild-type cells was measured before use. In the full gene space the offset between each screen's control cells and PD32 is 1.05 times the PD14–PD32 passage gap for the activation screen and 0.73 times for the interference screen, but nearly perpendicular to it (cosines +0.112 and +0.071 after matching sequencing depth; Supplementary Figure S2). In the 20-component space it is 0.84 and 0.65 times the gap, with cosines of −0.169 and −0.288, so it is somewhat further from perpendicular there. The direction from each screen's controls to PD14 therefore remains mostly the passage direction: cosine +0.645 (activation) and +0.793 (interference) in the full gene space, and +0.811 and +0.887 in the 20-component space. Both screens clear the pre-specified threshold of 0.70 in the 20-component space; in the full gene space the activation screen does not.

Supplementary Note S2. Removing cell division: the full result

To ask whether the passage signal can be separated from cell division, we removed a cell-division direction from the data and repeated the test. Three kinds of direction were fixed in advance for this test. The primary kind, shown in Figure 3B, is built from a screen's own control cells, so it comes in two versions, one per screen, each removed from that screen's data. The other two kinds, built from middle-passage PD26 cells and from the cell-cycle gene list, are secondary and are reported below. Each control-cell direction is built from one half of the screen's control cells, which never enter the starting profile, target or thresholds. In the 20-component space they lie 19.7° from the passage direction (activation screen; 20.6° for interference), so projecting them out removes 66.4% and 64.9% of the passage direction's length. In the full gene space the angles are 59.8° and 52.8°, removing 13.6% and 20.3%; angles from spaces of different dimension are not comparable, because two random directions sit about 80° apart on average in 20 dimensions and nearly 90° in 23,485. The adjusted test was not run in the full gene space, which cannot fire at these depths.

Removing a cell-division direction built from cells removes the passage signal with it (Figure 3B). From the own-control start, power to detect a planted half-gap effect falls from 0.895 with nothing removed (0.940 with all control cells as the start; Supplementary Figure S1B) to 0.010. A second direction fixed in advance, built from middle-passage PD26 wild-type cells that belong to neither screen and appear nowhere else in the analysis, lies 18.5° from passage and gives the same result from both starting points (power 0.040 from the own-control start and 0.715 from the late-passage start, against 0.010 and 0.760 for the control-cell direction), so the answer does not depend on the screens' engineered cells. By the pre-specified criteria, passage and cell division are inseparable in this dataset, and the adjusted test cannot detect a real effect. In this test the effects, the starting profile and the thresholds use only the half of the control cells not used to build the direction, so with nothing removed 82 perturbations are called rather than 80. Across the screen, removing the control-cell direction takes the called count from 82 to 27 (17 expected by chance) and leaves two of the four factors called. Because the same projection also removes a planted effect known to be real, these drops are not evidence either way. Removing the gene-list direction instead leaves 75 perturbations called (72 from the late-passage start) and all four factors called from both starts.

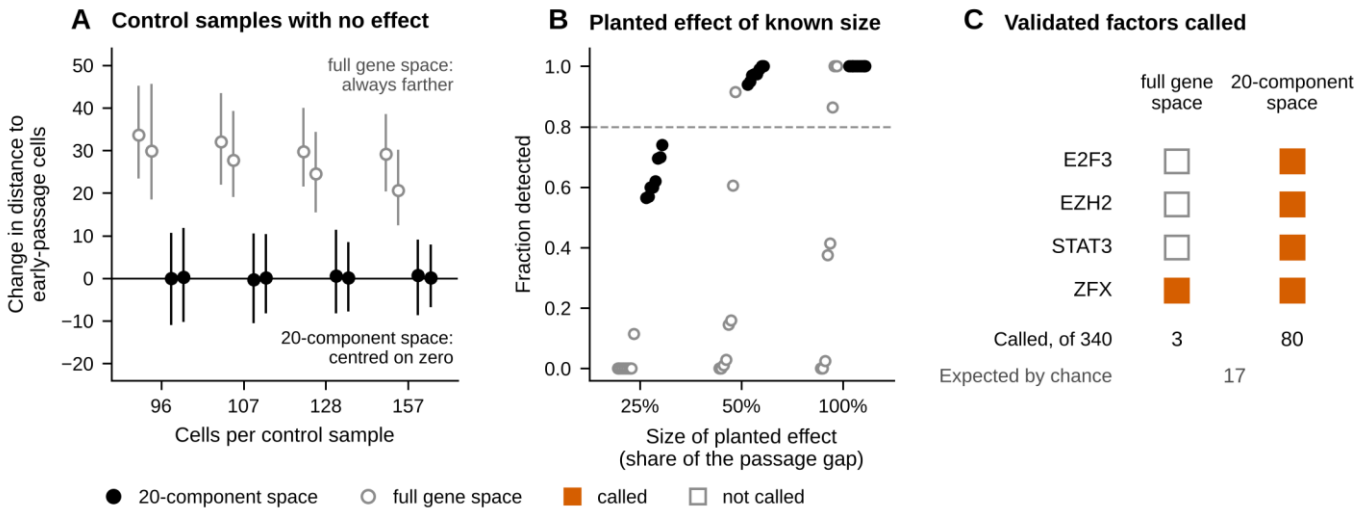
From the late-passage start, planted effects are still detected after removal with power 0.760 at worst, just short of the pre-specified 0.80 (Supplementary Figure S3D). This does not make passage and cell division separable: the removal is the same from either start and takes away the same two-thirds of the passage direction. The starts differ in how well the remaining third can be seen. A change in distance registers movement along the line from the start to the target; from late-passage cells that line is the passage direction itself, whereas from a screen's control cells it also carries the offset between the screen and the wild-type cells (Supplementary Figure S2A), a likely reason the remainder changes the distance so little. From the late-passage start the called count falls from 78 (73 with all control cells used for the effects and thresholds) to 9, below the 17 expected by chance, and none of the four factors remains called: what is left

of passage once cell division is removed shows no approach. Because power falls short of the criterion, this is descriptive.

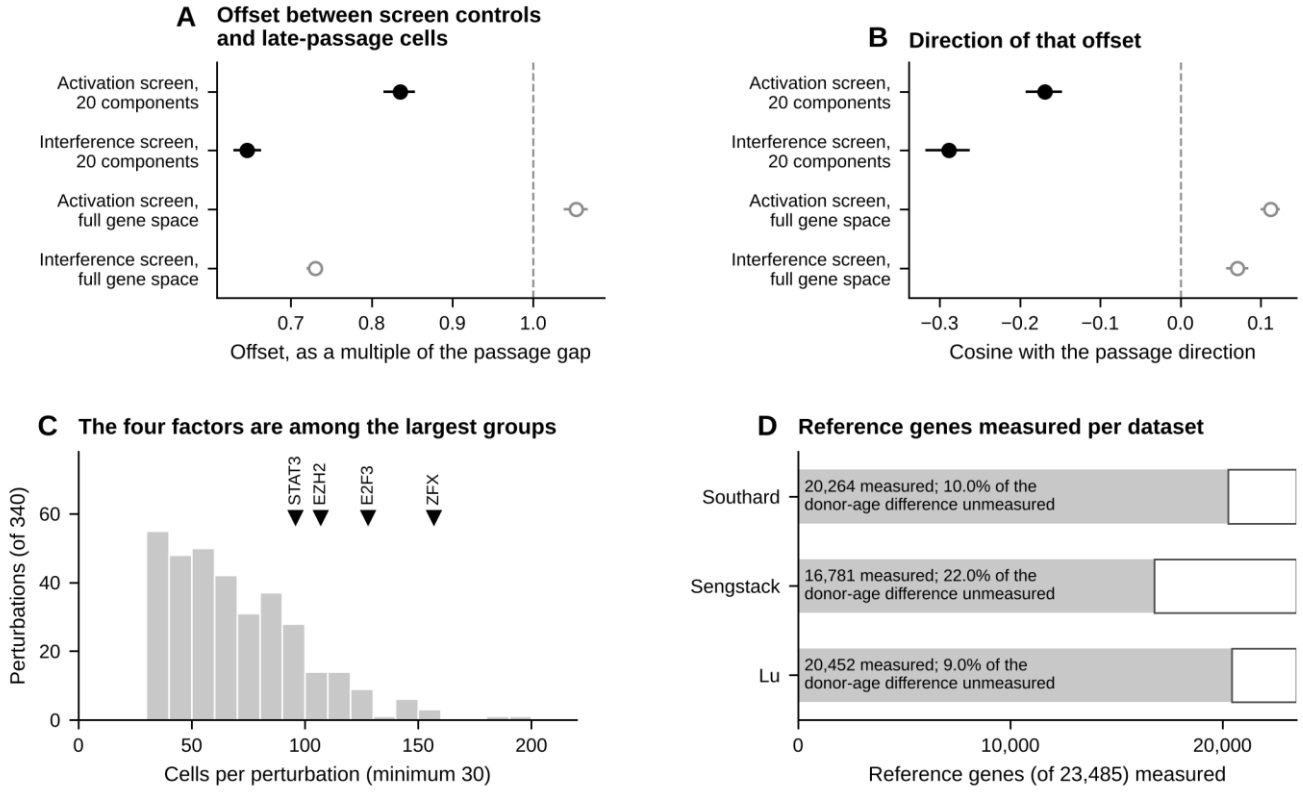
Supplementary Note S3. Cell-division directions not built from single cells

Two cell-division directions in Table 1 are not built from single cells. The gene-list direction lies 61.7° from passage and 48.8° from donor age, so with it, donor age rather than passage is the closer definition. The direction built from bulk GTEx samples gives 55.4° for passage and 35.8° for donor age. The bulk direction lies 50–53° from every direction built from cells, while those agree with one another within 12–36°. Passage lies 78.5° from donor age and 79.2° from the ruler in the 20-component space (87.7° and 88.6° in the full gene space), and the six pairs of aging definitions lie 70–90° apart in the 20-component space, so the aging definitions do not agree well with one another either.

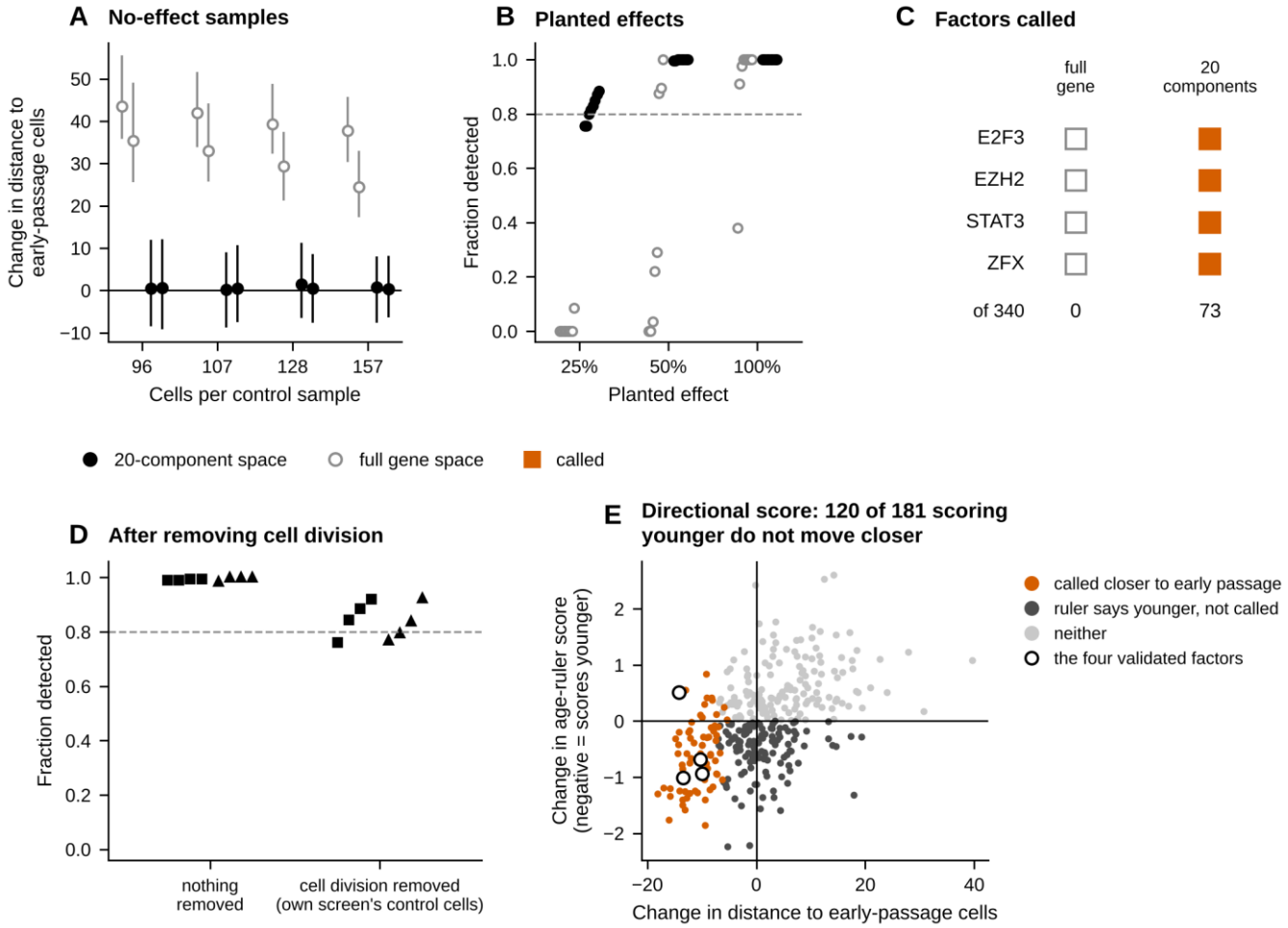
SUPPLEMENTARY FIGURES



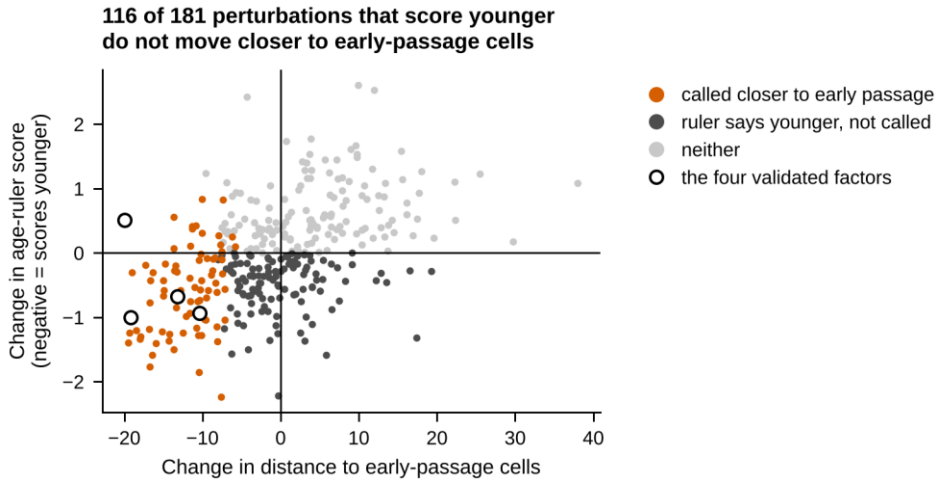
Supplementary Figure S1. The test works in the 20-component space but not in the full gene space. (A) Change in distance to early-passage cells for 200 random samples of control cells with no effect, at the group sizes of the four validated factors; point, median; bar, 95% range; one point per screen. In the full gene space (open) every sample moves farther away; in the 20-component space (filled) the spread is centered on zero. (B) Fraction of 200 control samples detected after a planted effect of 25, 50 or 100% of the passage gap; dashed line, 0.80. (C) The four validated factors in each space (filled, called), with the number called among all 340 perturbations. All panels use each perturbation's own control cells as the starting point.



Supplementary Figure S2. Structure of the data. (A) Offset between each screen's control cells and late-passage wild-type cells, as a multiple of the passage gap, and (B) its direction relative to the passage direction, after matching sequencing depth; median and 95% interval, 20-component space (filled) and full gene space (open). (C) Cells per perturbation for all 340 analyzed perturbations (167 activation, 173 interference); triangles mark the four validated factors. (D) Reference genes measured by each dataset, and the share of the GTEx donor-age difference that falls in unmeasured genes. Lu values are corrected for a gene-matching error (correction notice in the repository).



Supplementary Figure S3. Results from the late-passage starting point. The analyses of Supplementary Figures S1 and S4 and Figure 3B repeated with each effect added to late-passage (PD32) cells instead of the perturbation's own control cells. This start removes the offset between the screens and the wild-type cells (Supplementary Figure S2A). (A) Control samples with no effect. (B) Fraction of planted effects detected. (C) The four validated factors (filled, called) and the number called among 340. (D) Fraction of planted half-gap effects detected with nothing removed and after removing each screen's own control-cell direction; squares, activation screen; triangles, interference screen. (E) Directional age score against distance; of 181 perturbations that score younger, 120 are not called closer.



Supplementary Figure S4. A one-dimensional age score and distance disagree. Each perturbation's change in distance to early-passage cells against its change on the frozen age ruler of the companion preprint (negative, scores younger), from the perturbation's own control cells, 20-component space. Of 181 perturbations that score younger, 116 are not called closer to early-passage cells. Open circles, the four validated factors.